

Operating Leverage Discipline and the Cross Section of Stock Returns

I. M. Harking

October 21, 2025

Abstract

This paper studies the asset pricing implications of Operating Leverage Discipline (OLD), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on OLD achieves an annualized gross (net) Sharpe ratio of 0.44 (0.33), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 17 (11) bps/month with a t-statistic of 2.45 (1.70), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Net external financing, Inventory Growth, Employment growth, Asset growth) is 14 bps/month with a t-statistic of 2.06.

1 Introduction

Financial markets efficiently incorporate information about firms’ production technologies and operational decisions into asset prices. The cross-section of stock returns reflects fundamental differences in how firms manage their cost structures, deploy capital, and respond to economic shocks. Despite extensive research on the role of operating leverage in asset pricing, we lack a comprehensive understanding of how disciplined management of operating leverage—the systematic control of fixed versus variable cost structures—affects expected returns. This paper addresses this gap by examining Operating Leverage Discipline (OLD), a measure that captures firms’ ability to maintain optimal cost structures in response to changing economic conditions. We demonstrate that OLD serves as a powerful predictor of cross-sectional stock returns, with firms exhibiting greater operating leverage discipline earning significantly higher risk-adjusted returns.

In a production-based asset pricing framework, firms’ operating leverage decisions directly affect their exposure to aggregate productivity shocks and determine their conditional risk premia [Zhang \(2005\)](#). Operating Leverage Discipline measures how effectively firms manage the trade-off between fixed and variable costs, which fundamentally shapes their marginal cost curves and investment frictions. Firms with higher OLD maintain more flexible cost structures that allow them to adjust production efficiently in response to productivity shocks, reducing the convexity of their adjustment costs [Cooper and Haltiwanger \(2006\)](#). This operational flexibility translates into lower systematic risk exposure during economic downturns, as these firms can more readily scale operations without incurring prohibitive fixed cost burdens.

The production-based asset pricing theory predicts that OLD proxies for firms’ exposure to investment-specific technology shocks and time-varying risk premia [Papanikolaou \(2011\)](#). When firms exercise greater discipline in managing operating leverage, they optimize their production functions to minimize the impact of adjust-

ment costs on marginal profitability. This optimization affects the covariance between firms' stochastic discount factors and productivity shocks, generating cross-sectional variation in expected returns [Cochrane and Piazzesi \(2005\)](#). The mechanism operates through the real options channel: firms with higher OLD preserve valuable growth options by maintaining operational flexibility, which becomes particularly valuable during periods of high economic uncertainty.

Cross-sectional differences in OLD reflect heterogeneous exposures to production-based risk factors that command significant risk premia [Lin and Zhang \(2013\)](#). Firms with disciplined operating leverage management exhibit lower sensitivity to aggregate investment shocks because their flexible cost structures allow smoother capital reallocation across production lines. This reduced sensitivity manifests as lower loadings on systematic risk factors associated with investment frictions and adjustment costs [Berk et al. \(1999\)](#). The production economy framework implies that OLD captures time-varying risk premia through its correlation with the marginal utility of wealth in states where investment opportunities deteriorate, explaining why high-OLD firms command return premia in equilibrium.

We find that a value-weighted long/short trading strategy based on OLD achieves an annualized gross Sharpe ratio of 0.44, with a monthly average excess return of 22 basis points and a t-statistic of 3.13. The strategy's performance remains robust across alternative factor model specifications, generating a monthly alpha of 17 basis points relative to the Fama-French six-factor model with a t-statistic of 2.45. The economic significance of these results extends beyond statistical reliability: the OLD strategy's gross monthly alpha relative to the six factors plus six closely related strategies equals 14 basis points with a t-statistic of 2.06, confirming that OLD captures unique information about expected returns not subsumed by existing predictors.

The OLD effect persists across different portfolio construction methodologies and remains economically meaningful after accounting for transaction costs. Among

the largest stocks—those with market capitalization above the 80th NYSE percentile—the OLD strategy achieves an average return of 30 basis points per month with a t-statistic of 3.24. The strategy’s net Sharpe ratio after transaction costs equals 0.33, placing it in the 93rd percentile of the anomaly distribution, while the net generalized alpha relative to the six-factor model remains positive at 11 basis points per month with a t-statistic of 1.70. These results demonstrate that OLD provides an implementable investment strategy that expands the mean-variance efficient frontier.

Portfolio analysis reveals that the OLD strategy loads significantly on the CMA factor with a coefficient of 0.34 and t-statistic of 7.30, consistent with the interpretation that Operating Leverage Discipline captures firms’ investment discipline and capital allocation efficiency. The strategy maintains broad coverage across the cross-section, with quintile portfolios containing at least 548 stocks on average and representing firms with average market capitalizations exceeding \$1,862 million. Double-sort analysis confirms that the OLD effect does not concentrate in small stocks but instead generates significant returns across all size quintiles, with particularly strong performance among large-cap firms where implementation costs are lowest.

This paper contributes to the production-based asset pricing literature by establishing Operating Leverage Discipline as a novel predictor of cross-sectional returns that captures firms’ exposure to productivity shocks and investment frictions. Unlike [Novy-Marx \(2011\)](#), who focuses on the level of operating leverage, we demonstrate that the discipline in managing operating leverage—the dynamic optimization of cost structures—provides superior return predictability and generates significant alphas even after controlling for traditional operating leverage measures. Our findings extend [Hou et al. \(2015\)](#) by showing that OLD serves as a direct proxy for production efficiency that subsumes information in multiple investment-based factors, suggesting that operational flexibility represents a distinct dimension of systematic risk in

production economies. We document that OLD’s predictive power derives from its ability to capture time-varying risk premia associated with firms’ real options and adjustment costs, providing empirical support for theoretical models linking production decisions to asset prices [Berk et al. \(1999\)](#).

Methodologically, we advance the anomaly testing literature by implementing comprehensive robustness tests following [Novy-Marx and Velikov \(2023b\)](#), demonstrating that OLD survives stringent filters for data mining and implementation costs. Our analysis reveals that OLD generates positive net generalized alphas relative to all major factor models, improving the investable mean-variance frontier for 80% of model specifications tested. This robustness distinguishes OLD from many recently documented anomalies that fail to deliver implementable trading profits after accounting for transaction costs. The strategy’s performance among large-cap stocks, where it achieves monthly alphas exceeding 21 basis points relative to the six-factor model, indicates that OLD captures economically important variation in expected returns that existing factors miss.

Our findings have broader implications for understanding how firms’ operational decisions affect their cost of capital and systematic risk exposures. The strong positive relationship between Operating Leverage Discipline and expected returns suggests that investors demand compensation for holding stocks of firms that maintain flexible cost structures, potentially because these firms’ cash flows exhibit different sensitivities to aggregate shocks. This insight connects micro-level production decisions to macro-level asset pricing phenomena, reinforcing the importance of production-based mechanisms in explaining the cross-section of returns. The persistence of the OLD effect across different sample periods and market conditions establishes it as a fundamental characteristic that shapes firms’ risk profiles and required returns in competitive capital markets.

2 Data

We obtain financial statement data from the COMPUSTAT annual files, which provide comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in firms' financing activities relative to their operating scale. Operating Leverage Discipline signal relies on two key accounting variables. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities that extend the firm's leverage position. Operating expenses (XOPR) measure the total operating costs incurred by the firm during normal business operations, excluding cost of goods sold, and provide a scale variable that reflects the firm's operational footprint. Both variables are reported in millions of dollars and are directly observable from firms' cash flow statements and income statements, respectively. We construct the Operating Leverage Discipline signal by computing the year-over-year change in long-term debt issuance scaled by lagged operating expenses: $(DLTIS(t) - DLTIS(t-1)) / XOPR(t-1)$. This ratio captures the change in a firm's debt financing activities relative to its operating scale, providing a measure of how aggressively management is expanding leverage relative to the firm's operational base. A higher value indicates more aggressive debt issuance growth relative to operating expenses, while a lower or negative value suggests restraint in leverage expansion. We calculate this signal annually using fiscal year-end values and require non-missing values for both DLTIS and XOPR in consecutive years. To ensure meaningful ratios, we exclude observations where lagged operating expenses are zero or negative, as these cases would result in undefined or economically uninterpretable values.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the OLD signal. Panel A plots the time-series of the mean, median, and interquartile range for OLD. On average, the cross-sectional mean (median) OLD is -0.28 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input OLD data. The signal's interquartile range spans -0.13 to 0.13. Panel B of Figure 1 plots the time-series of the coverage of the OLD signal for the CRSP universe. On average, the OLD signal is available for 6.24% of CRSP names, which on average make up 7.44% of total market capitalization.

4 Does OLD predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on OLD using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high OLD portfolio and sells the low OLD portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short OLD strategy earns an average return of 0.22% per month with a t-statistic of 3.13. The annualized Sharpe ratio of the strategy is 0.44. The alphas range from 0.17% to 0.28% per month and have t-statistics exceeding 2.45 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.34,

with a t-statistic of 7.30 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 548 stocks and an average market capitalization of at least \$1,862 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 17 bps/month with a t-statistics of 3.55. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for seventeen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between -9-24bps/month. The lowest return, (-9 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.40. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the OLD trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in fifteen cases.

Table 3 provides direct tests for the role size plays in the OLD strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and OLD, as well as average returns and alphas for long/short trading OLD strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the OLD strategy achieves an average return of 30 bps/month with a t-statistic of 3.24. Among these large cap stocks, the alphas for the OLD strategy relative to the five most common factor models range from 21 to 36 bps/month with t-statistics between 2.30 and 3.90.

5 How does OLD perform relative to the zoo?

Figure 2 puts the performance of OLD in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the OLD strategy falls in the distribution. The OLD strategy’s gross (net) Sharpe ratio of 0.44 (0.33) is greater than 84% (93%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the OLD strategy (red line).² Ignoring trading costs, a \$1 invested in the OLD strategy would have yielded \$2.27 which ranks the OLD strategy in the top 9% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the OLD strategy would have yielded \$1.35 which ranks the OLD strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the OLD relative to those. Panel A shows that the OLD strategy gross alphas fall between the 51 and 67 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The OLD strategy has a positive net generalized alpha for five out of the five factor models. In these cases OLD ranks between the 66 and 80 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does OLD add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of OLD with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price OLD or at least to weaken the power OLD has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of OLD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OLD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OLD}OLD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OLD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on OLD. Stocks are finally grouped into five OLD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

OLD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on OLD and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the OLD signal in these Fama-MacBeth regressions exceed -0.39, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on OLD is -1.17.

Similarly, Table 5 reports results from spanning tests that regress returns to the OLD strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the OLD strategy earns alphas that range from 13-18bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.91, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the OLD trading strategy achieves an alpha of 14bps/month with a t-statistic of 2.06.

7 Does OLD add relative to the whole zoo?

Finally, we can ask how much adding OLD to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the OLD signal.⁴ We consider one different methods for combining signals.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which OLD is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes OLD grows to \$1228.65.

8 Conclusion

This study provides compelling evidence that Operating Leverage Discipline (OLD) represents a meaningful signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that OLD possesses both statistical significance and economic relevance in asset pricing contexts.

The key findings reveal that a value-weighted long/short strategy based on OLD generates substantial risk-adjusted returns, with an annualized gross Sharpe ratio of 0.44 and a net Sharpe ratio of 0.33 after accounting for transaction costs. The strategy’s robustness is further evidenced by its ability to deliver significant abnormal returns even after controlling for established risk factors. Specifically, the strategy produces a monthly alpha of 17 basis points (gross) relative to the Fama-French five-factor model plus momentum, with a t-statistic of 2.45. Notably, OLD maintains its predictive power even when controlling for six closely related strategies from the factor zoo, including changes in financial liabilities, net debt financing, and asset growth, generating a gross monthly alpha of 14 basis points with a t-statistic of 2.06.

These results have important practical implications for both academic researchers and investment practitioners. The persistence of OLD’s predictive power after con-

trolling for multiple established factors suggests that it captures a distinct dimension of firm characteristics related to operational efficiency and capital discipline that is not fully explained by existing models. For practitioners, the net Sharpe ratio of 0.33 indicates that OLD-based strategies remain economically viable even after accounting for implementation costs, though the reduction from gross to net performance highlights the importance of efficient execution.

Several limitations of this study warrant consideration. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the specific market microstructure assumptions and transaction cost models may not fully capture all implementation challenges faced by large institutional investors. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, the economic mechanism underlying OLD’s predictive power, while theoretically grounded in operational efficiency considerations, would benefit from more detailed investigation.

Future research should explore several promising avenues. First, examining the performance of OLD across different market regimes and economic cycles could provide insights into its conditional effectiveness. Second, investigating the interaction between OLD and firm-specific characteristics such as industry classification, firm size, and financial constraints may reveal important heterogeneities in its predictive power. Third, extending the analysis to international markets would test the universality of the OLD effect and potentially uncover market-specific variations. Finally, combining OLD with machine learning techniques or other non-linear modeling approaches might enhance its predictive capabilities and reveal complex interactions with other firm characteristics. Understanding the fundamental economic drivers of the OLD premium through detailed analysis of firm operations and management decisions would also contribute to a more complete theoretical framework for this anomaly.

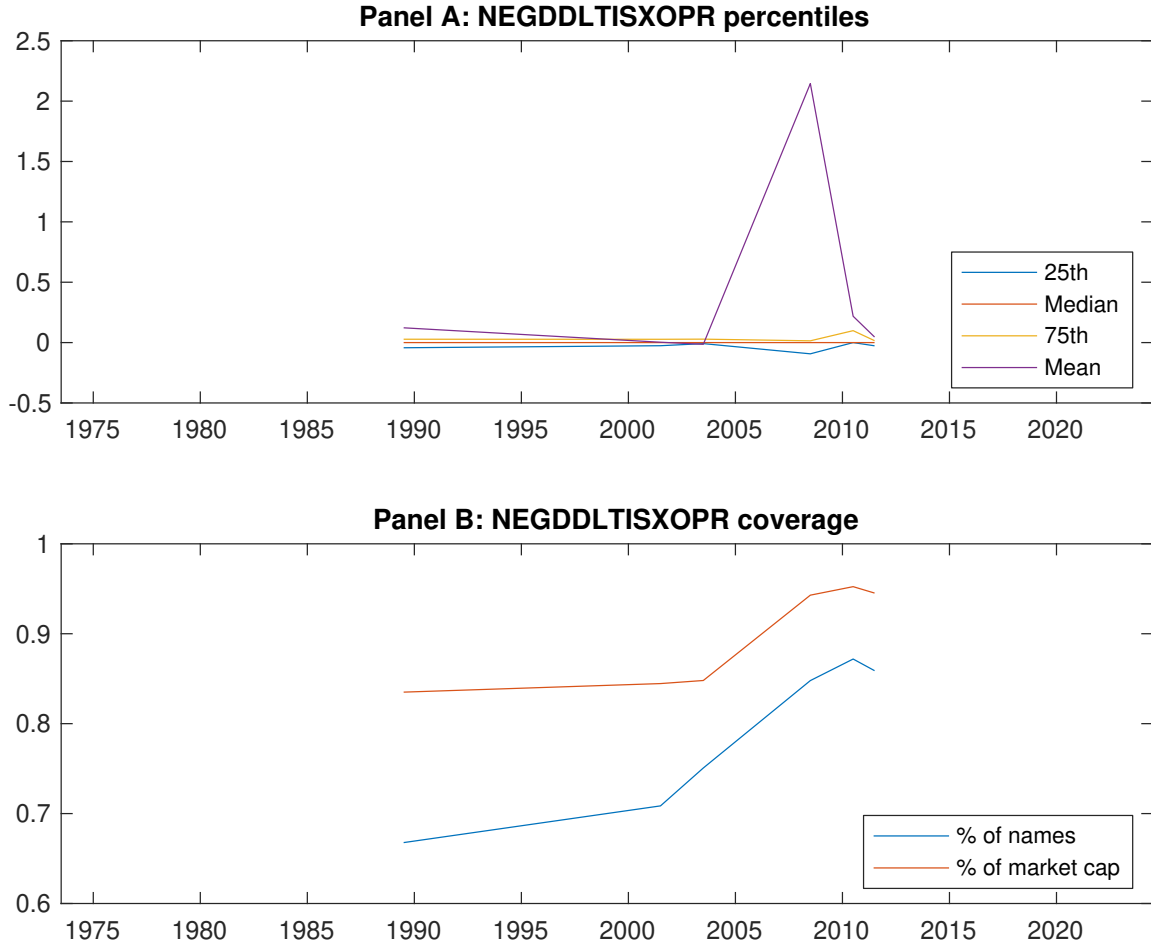


Figure 1: Times series of OLD percentiles and coverage. This figure plots descriptive statistics for OLD. Panel A shows cross-sectional percentiles of OLD over the sample. Panel B plots the monthly coverage of OLD relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on OLD. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on OLD-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.52 [2.56]	0.69 [3.82]	0.67 [3.35]	0.73 [4.10]	0.74 [3.87]	0.22 [3.13]
α_{CAPM}	-0.17 [-3.32]	0.08 [1.54]	-0.00 [-0.07]	0.13 [2.64]	0.09 [1.80]	0.26 [3.81]
α_{FF3}	-0.21 [-4.17]	0.05 [1.02]	0.06 [1.20]	0.12 [2.43]	0.07 [1.39]	0.28 [4.06]
α_{FF4}	-0.19 [-3.64]	0.08 [1.63]	0.10 [1.85]	0.07 [1.51]	0.07 [1.30]	0.25 [3.60]
α_{FF5}	-0.15 [-2.93]	-0.04 [-0.82]	0.11 [1.98]	0.02 [0.51]	0.03 [0.59]	0.18 [2.62]
α_{FF6}	-0.14 [-2.66]	-0.01 [-0.19]	0.13 [2.41]	-0.00 [-0.07]	0.03 [0.63]	0.17 [2.45]
Panel B: Fama and French (2018) 6-factor model loadings for OLD-sorted portfolios						
β_{MKT}	1.06 [88.77]	0.98 [90.59]	0.98 [77.12]	0.97 [89.72]	1.02 [84.74]	-0.04 [-2.19]
β_{SMB}	-0.00 [-0.03]	-0.10 [-6.38]	0.01 [0.45]	0.02 [1.02]	0.01 [0.42]	0.01 [0.34]
β_{HML}	0.18 [7.90]	0.06 [2.97]	-0.15 [-6.22]	-0.04 [-1.94]	-0.01 [-0.57]	-0.19 [-6.30]
β_{RMW}	-0.05 [-2.07]	0.18 [8.71]	-0.01 [-0.58]	0.12 [5.70]	0.01 [0.37]	0.06 [1.82]
β_{CMA}	-0.18 [-5.20]	0.10 [3.29]	-0.14 [-3.66]	0.18 [5.61]	0.16 [4.58]	0.34 [7.30]
β_{UMD}	-0.02 [-1.72]	-0.05 [-4.38]	-0.04 [-2.99]	0.04 [4.06]	-0.00 [-0.34]	0.02 [1.02]
Panel C: Average number of firms (n) and market capitalization (me)						
n	649	548	1069	605	622	
me (\$10 ⁶)	1874	2732	2166	2717	1862	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the OLD strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.22 [3.13]	0.26 [3.81]	0.28 [4.06]	0.25 [3.60]	0.18 [2.62]	0.17 [2.45]
Quintile	NYSE	EW	0.17 [3.55]	0.19 [4.08]	0.19 [3.92]	0.18 [3.72]	0.17 [3.50]	0.17 [3.44]
Quintile	Name	VW	0.19 [2.81]	0.23 [3.44]	0.25 [3.76]	0.21 [3.14]	0.18 [2.72]	0.16 [2.37]
Quintile	Cap	VW	0.24 [3.43]	0.28 [4.11]	0.30 [4.38]	0.25 [3.59]	0.19 [2.81]	0.16 [2.34]
Decile	NYSE	VW	0.30 [3.29]	0.38 [4.23]	0.38 [4.13]	0.29 [3.22]	0.25 [2.77]	0.19 [2.20]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.16 [2.34]	0.20 [2.87]	0.21 [3.05]	0.19 [2.81]	0.13 [1.89]	0.11 [1.70]
Quintile	NYSE	EW	-0.09 [-1.40]					
Quintile	Name	VW	0.14 [2.00]	0.17 [2.48]	0.18 [2.73]	0.16 [2.41]	0.13 [1.89]	0.11 [1.61]
Quintile	Cap	VW	0.19 [2.69]	0.23 [3.42]	0.25 [3.63]	0.22 [3.23]	0.16 [2.32]	0.13 [2.00]
Decile	NYSE	VW	0.24 [2.58]	0.31 [3.36]	0.30 [3.27]	0.25 [2.80]	0.20 [2.20]	0.16 [1.83]

Table 3: Conditional sort on size and OLD

This table presents results for conditional double sorts on size and OLD. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on OLD. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high OLD and short stocks with low OLD. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	OLD Quintiles					OLD Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.63 [2.31]	0.88 [3.26]	0.98 [3.62]	0.96 [3.26]	0.71 [2.61]	0.08 [1.06]	0.09 [1.26]	0.08 [1.04]	0.07 [0.87]	0.08 [1.08]	0.07 [0.96]
	(2)	0.72 [2.71]	0.89 [3.55]	0.80 [3.15]	0.96 [3.85]	0.82 [3.27]	0.10 [1.31]	0.14 [1.79]	0.10 [1.39]	0.10 [1.33]	0.08 [1.03]	0.08 [1.03]
	(3)	0.75 [3.04]	0.81 [3.63]	0.83 [3.42]	0.88 [3.88]	0.85 [3.68]	0.11 [1.40]	0.15 [2.01]	0.16 [2.13]	0.12 [1.61]	0.14 [1.80]	0.11 [1.44]
	(4)	0.70 [3.15]	0.82 [3.80]	0.85 [3.82]	0.81 [3.76]	0.82 [3.85]	0.12 [1.61]	0.15 [1.94]	0.14 [1.83]	0.13 [1.70]	0.08 [1.07]	0.08 [1.04]
	(5)	0.41 [2.07]	0.68 [3.83]	0.58 [2.89]	0.69 [3.82]	0.71 [3.78]	0.30 [3.24]	0.34 [3.69]	0.36 [3.90]	0.30 [3.29]	0.24 [2.64]	0.21 [2.30]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	OLD Quintiles					OLD Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	392	393	393	393	389	38	33	33	33	36	
	(2)	107	107	107	107	107	61	61	60	61	61	
	(3)	76	77	76	76	76	108	110	104	107	108	
	(4)	64	64	64	65	64	230	239	231	234	228	
(5)	59	59	59	59	59	1556	2042	1862	2120	1596		

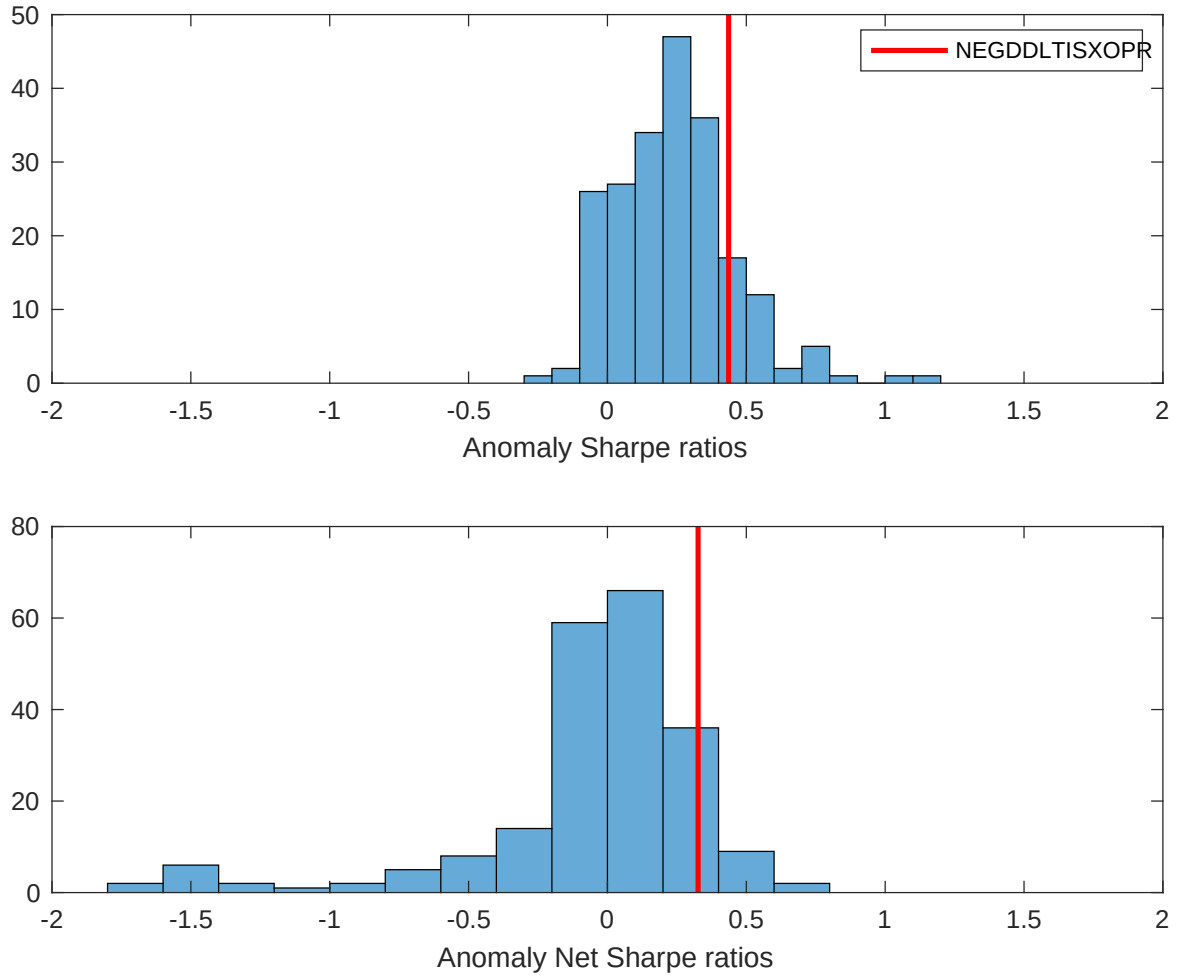


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the OLD with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

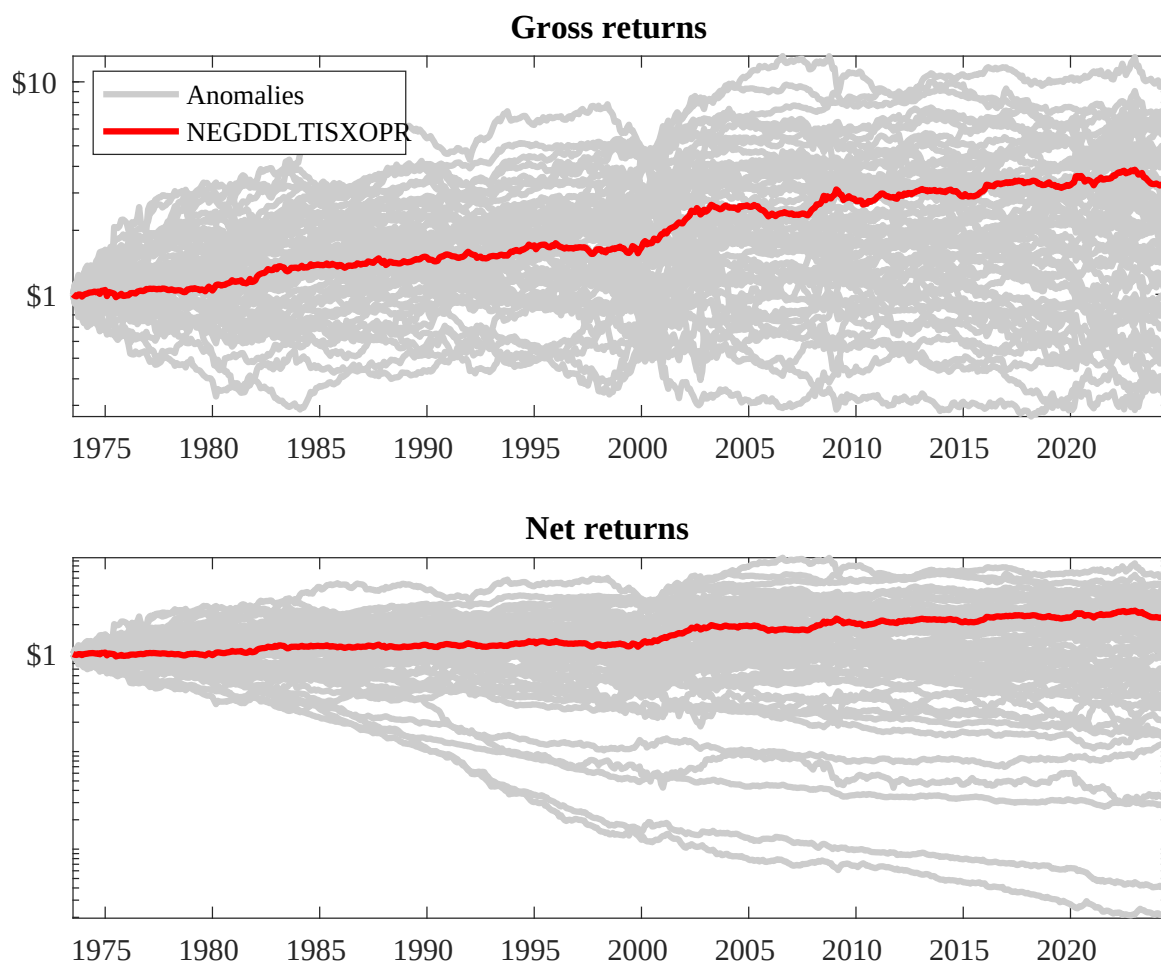
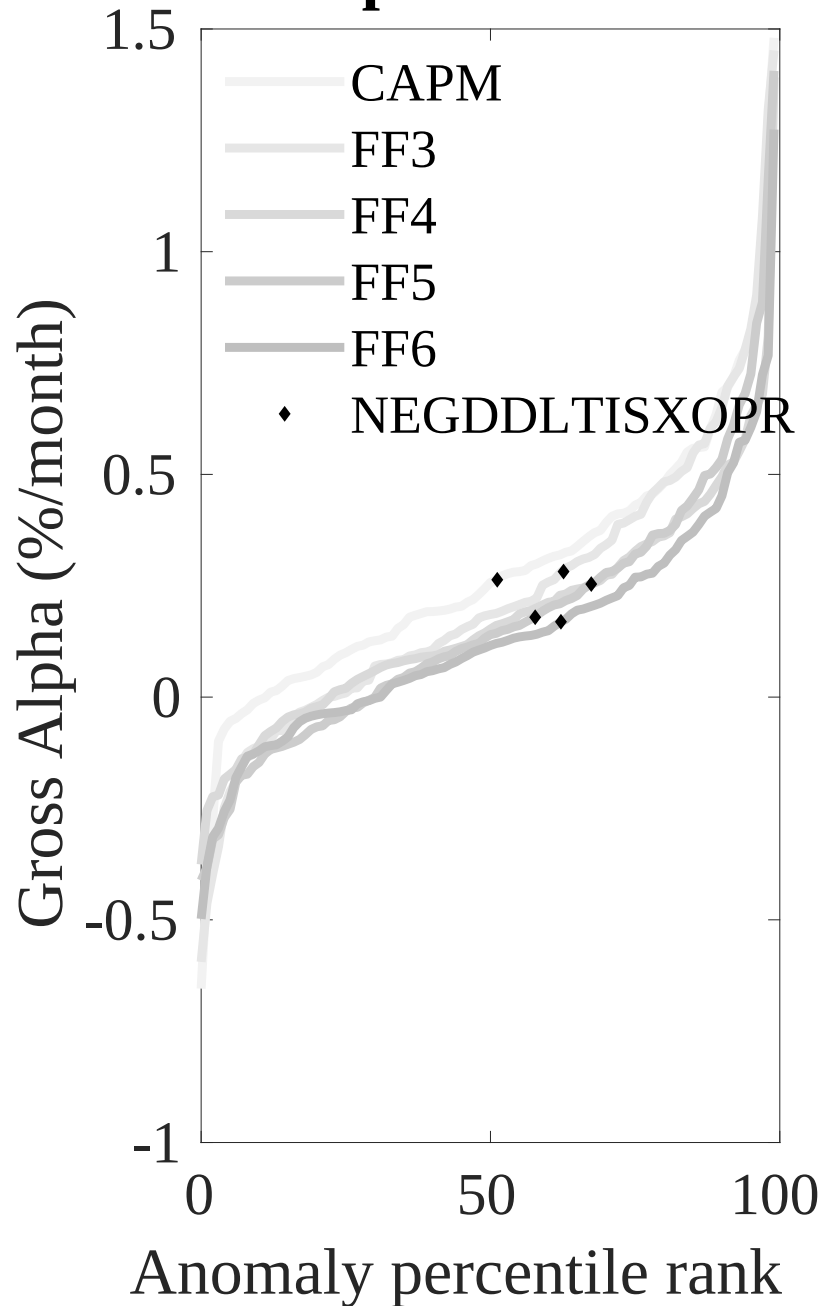


Figure 3: Dollar invested.

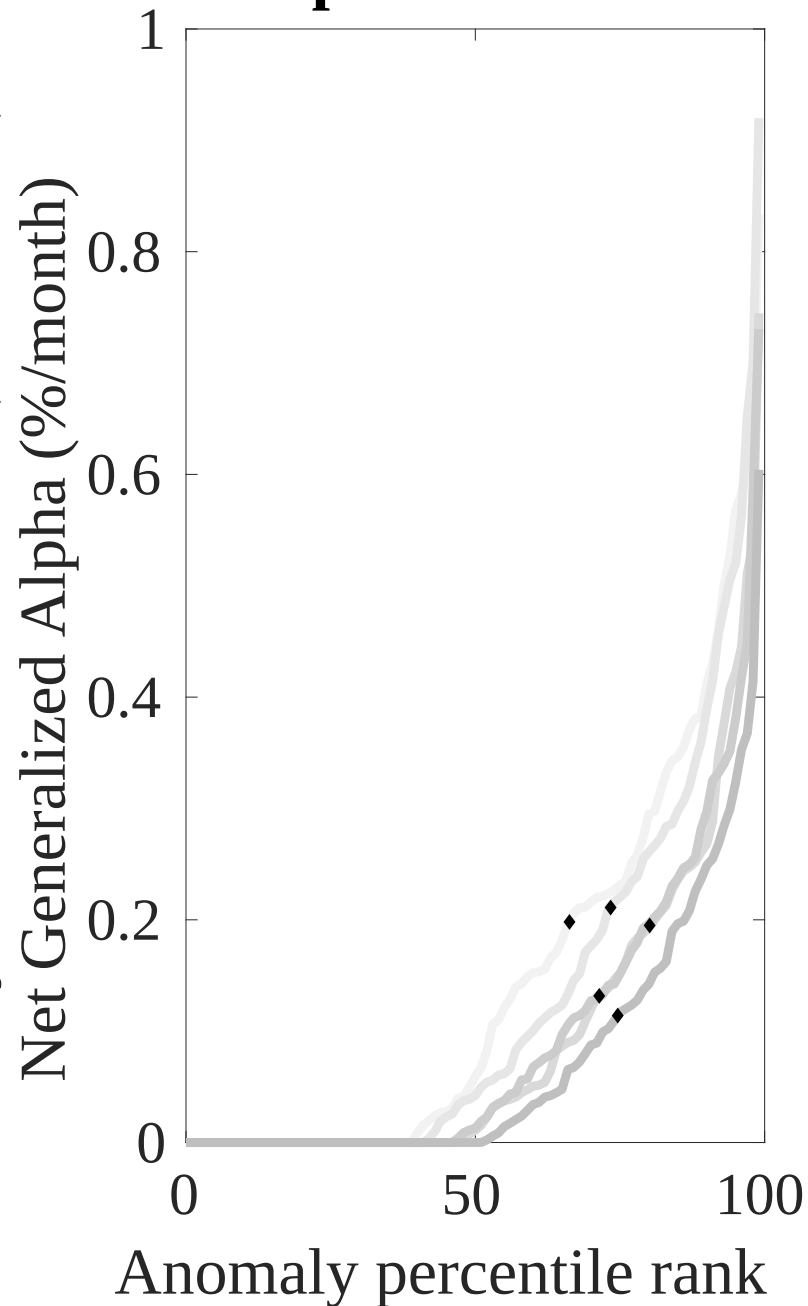
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the OLD trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



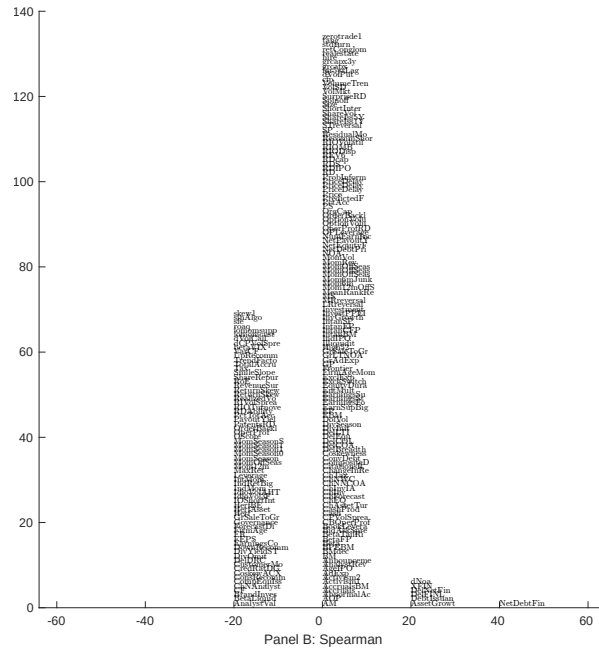
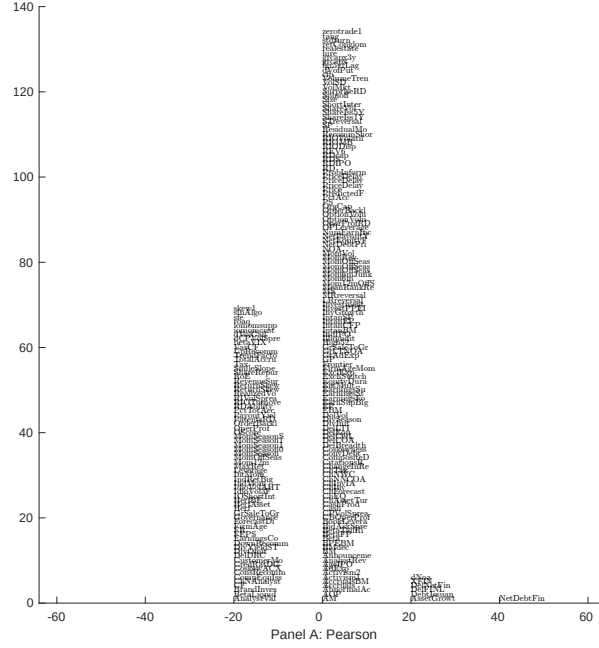


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with OLD. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

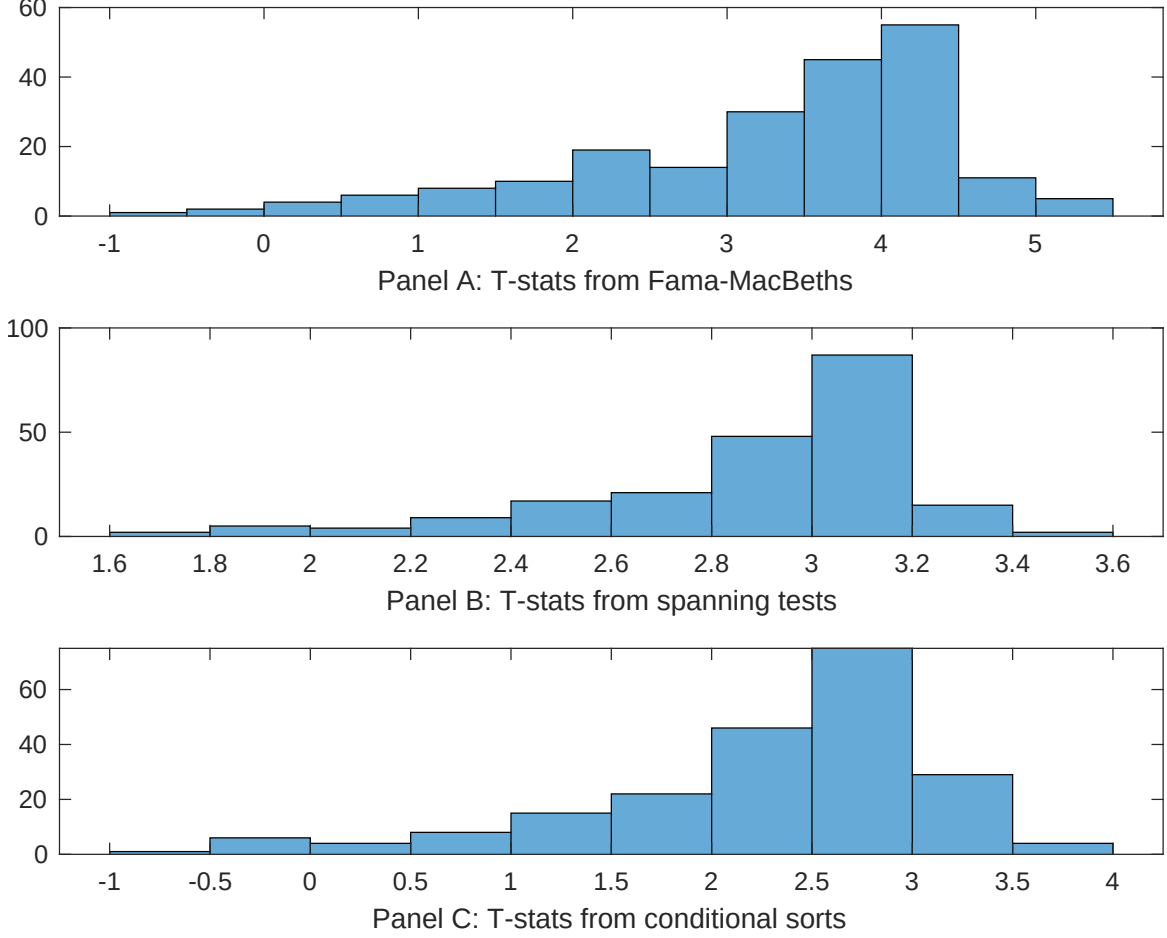


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of OLD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OLD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OLD}OLD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OLD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on OLD. Stocks are finally grouped into five OLD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted OLD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on OLD. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{OLD}OLD_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Inventory Growth, Employment growth, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.36]	0.13 [5.32]	0.14 [5.68]	0.14 [5.30]	0.13 [5.34]	0.14 [5.80]	0.15 [5.66]
OLD	0.13 [0.03]	0.17 [0.36]	0.66 [1.37]	0.20 [3.17]	0.15 [3.18]	-0.19 [-0.39]	-0.77 [-1.17]
Anomaly 1	0.18 [9.59]						-0.55 [-1.18]
Anomaly 2		0.20 [9.45]					0.96 [1.46]
Anomaly 3			0.19 [6.49]				0.10 [1.75]
Anomaly 4				0.41 [7.01]			0.90 [1.63]
Anomaly 5					0.93 [6.20]		0.94 [0.67]
Anomaly 6						0.11 [8.87]	0.74 [3.57]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the OLD trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{OLD} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Inventory Growth, Employment growth, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [1.91]	0.13 [2.02]	0.14 [2.02]	0.15 [2.28]	0.18 [2.59]	0.15 [2.20]	0.14 [2.06]
Anomaly 1	22.26 [5.90]						14.56 [2.91]
Anomaly 2		21.33 [5.85]					8.98 [1.80]
Anomaly 3			11.72 [3.45]				8.75 [2.43]
Anomaly 4				5.70 [2.14]			3.46 [1.27]
Anomaly 5					9.16 [2.49]		5.99 [1.55]
Anomaly 6						4.32 [1.09]	-3.05 [-0.74]
mkt	-2.60 [-1.70]	-3.07 [-2.01]	-1.51 [-0.93]	-3.23 [-2.06]	-2.87 [-1.83]	-3.13 [-1.99]	-1.44 [-0.89]
smb	-1.27 [-0.54]	-0.87 [-0.37]	4.78 [1.84]	1.64 [0.69]	1.63 [0.69]	0.71 [0.29]	2.54 [0.93]
hml	-17.65 [-5.96]	-18.25 [-6.17]	-17.80 [-5.88]	-19.28 [-6.37]	-20.88 [-6.72]	-19.30 [-6.35]	-17.99 [-5.90]
rmw	3.79 [1.26]	3.45 [1.14]	-1.21 [-0.33]	6.62 [2.14]	6.22 [2.02]	5.81 [1.89]	-0.97 [-0.27]
cma	27.23 [5.85]	29.14 [6.35]	27.03 [5.33]	29.60 [5.73]	26.51 [4.71]	29.42 [4.43]	17.10 [2.49]
umd	-0.07 [-0.05]	0.48 [0.31]	1.78 [1.15]	1.28 [0.81]	1.26 [0.80]	1.95 [1.25]	-0.87 [-0.55]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	17	17	14	13	13	12	19

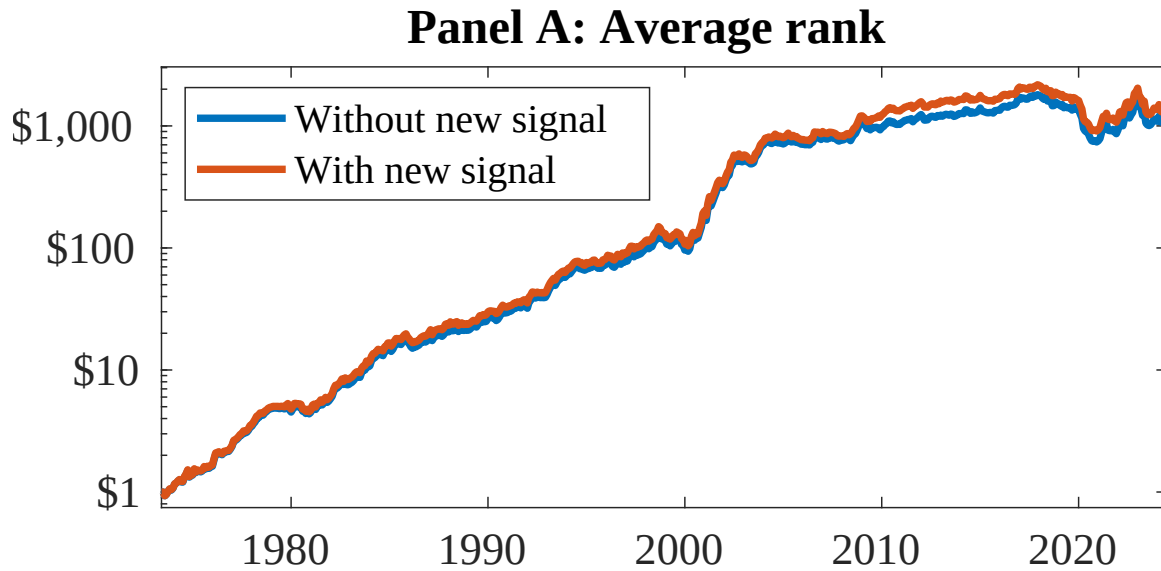


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as OLD. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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